

Subject: Operating Systems

Module: Process Management

A process is a program in execution. It includes the program counter, process state, CPU registers, and memory allocated to the process. The operating system is responsible for managing processes efficiently. **Process States**

A process can exist in different states such as New, Ready, Running, Waiting, and Terminated. The process state indicates the current activity of a process. **Process Control Block (PCB)**

The Process Control Block is a data structure maintained by the operating system for each process. It contains information such as process ID, process state, CPU registers, and scheduling information. **CPU Scheduling**

CPU scheduling is the process of deciding which process will use the CPU at a given time.

Common CPU scheduling algorithms include First Come First Serve (FCFS), Shortest Job First (SJF), and Round Robin. **Context Switching**

Context switching is the mechanism of saving the state of a currently running process and loading the state of the next process. It allows multiple processes to share a single CPU.